

Effect of cholesterol crystals on plaques and intima in arteries of patients with acute coronary and cerebrovascular syndromes.

.

Plaque disruption (PD) causes most acute cardiovascular events. Although cholesterol crystals (CCs) have been observed in plaques, their role in PD was unknown. However, cholesterol expands with crystallization tearing and perforating fibrous tissues. This study tested the hypothesis that CCs can damage plaques and intima, triggering PD, as observed in tissues prepared without ethanol solvents that dissolve CCs. Coronary arteries of patients who died of acute coronary syndrome (n = 19) and non-acute coronary syndrome causes (n = 12) and carotid plaques from patients with (n = 51) and without (n = 19) neurologic symptoms were studied. Samples were examined for CCs perforating the intima using light and scanning electron microscopy (SEM) with ethanol or vacuum dehydration. In addition, fresh unfixed carotid plaques were examined at 37 degrees C using confocal microscopy. Crystal content using SEM was scored from 0 to +3. SEM using vacuum dehydration had significantly higher crystal content compared with SEM using ethanol dehydration (+2.5 +/- 0.53 vs +0.25 +/- 0.46; p <0.0003), with enhanced detection of CC perforations. The presence of CCs using SEM and confocal microscopy was similar, suggesting that CC perforation can occur in vivo at 37 degrees C. All patients with acute coronary syndrome had perforating CCs, but none was present in patients without acute coronary syndrome (p = 0.0001). For all plaques, there were strong associations of CCs with PD, thrombus, symptoms (p <0.0001), and plaque size (p <0.02). Crystal content was an independent predictor of thrombus and symptoms. In conclusion, by avoiding ethanol in tissue preparation, CCs perforating the intima were shown to be associated with PD. Crystal content was significantly associated with clinical events, suggesting that cholesterol crystallization may have a role in PD.